MACHINE LEARNING – INNOVATIVE ASSIGNMENT
IA Guidelines & Requirements

The objective is to move beyond simply applying an ML algorithm and demonstrate a complete, justified, reproducible and usable ML solution.

1) Purpose of the Innovative Assignment

Students may select any meaningful problem from a suitable application domain and develop an end-to-end Machine Learning solution. The project may use classical ML, advanced ML, ensemble learning, or Deep Learning where justified by the problem. The emphasis is on the complete problem-solving pipeline rather than only on model accuracy.

Possible domains include:

- Healthcare and medical analytics
- Finance, banking and fraud detection
- Cybersecurity and intrusion/anomaly detection
- Agriculture and environmental monitoring
- Education and student analytics
- Retail, recommendation and customer analytics
- Transportation, IoT and smart systems
- Natural Language Processing, computer vision and speech
- Manufacturing, predictive maintenance and quality control
- Any other domain where ML provides a meaningful solution

2) Mandatory End-to-End Project Pipeline

Every project should follow the pipeline below. Some stages may be adapted to the problem, but students must explicitly state whether a stage is applicable and justify its inclusion or exclusion.

Stage	Pipeline Component	What Students Must Demonstrate
1	**Problem Definition & Use Case**	Define the real-world problem, users/stakeholders, objective, inputs, outputs, constraints and expected impact.
2	**Data Acquisition & Understanding**	Identify data source, data ownership/licence, dataset size, features, target variable, data types and limitations.
3	**Data Preparation**	Integrate/collect data, remove obvious errors, resolve duplicates, correct formats and prepare a usable dataset.
4	**Data Preprocessing**	Handle missing values, outliers, inconsistent values, categorical variables, scaling/normalization and other problem-specific transformations.
5	**Data Balancing**	Check class distribution. If imbalance exists, justify methods such as class weights, undersampling, oversampling or SMOTE and avoid data leakage.
6	**Exploratory Data Analysis (EDA)**	Use statistical summaries and suitable visualizations to understand distributions, relationships, trends, correlations and anomalies.
7	**Feature Engineering / Selection**	Create, transform, encode or select informative features. Explain why the selected features are relevant.
8	**Baseline Model**	Develop a simple baseline model to establish a meaningful reference point before advanced modelling.
9	**ML / Advanced ML / DL Modelling**	Select suitable algorithms and justify the choice. Compare multiple relevant approaches where appropriate.
10	**Hyperparameter Tuning**	Use suitable validation/tuning methods such as Grid Search, Random Search or other justified approaches.

11	**Validation & Evaluation**	Evaluate using metrics appropriate to the problem. Use train/validation/test separation or cross-validation as applicable.
12	**Use Case, Impact & Limitations**	Explain who benefits, how the system would be used, practical impact, risks, limitations and possible future improvements.

3) Core Principle: Every Major Decision Must Be Justified

Students should not treat the project as a sequence of library calls. For every important technical decision, answer: What was done? Why was it done? How was it implemented? What changed because of it? Was the decision validated?

- Why was this dataset selected?
- Why were particular preprocessing techniques used?
- Why was balancing required or not required?
- Why were these features selected/engineered?
- Why were these ML/DL algorithms selected?
- Why were these evaluation metrics selected?
- Why is the proposed model better or more useful than the baseline?
- What are the major errors/failure cases?
- How would the solution work with new/unseen data?

4) Data Preparation, Preprocessing & Balancing – Mandatory Checks

- Dataset source, collection method and licence/usage conditions
- Number of records and features before and after preprocessing
- Target variable and class distribution
- Missing-value analysis and treatment
- Duplicate-record analysis
- Outlier analysis and treatment where appropriate
- Categorical-variable encoding
- Scaling/normalization where required
- Data leakage check
- Class-imbalance analysis and balancing strategy, if applicable
- Train/validation/test split strategy
- Reproducibility through fixed random seeds where appropriate

Important: Balancing must be performed carefully. Students must avoid allowing information from the validation/test set to influence training or resampling decisions. The final test set should represent unseen data.

5) EDA Requirements

EDA should be analytical, not decorative. Include only plots/tables that help answer meaningful questions.

- Univariate analysis: distributions, frequencies, descriptive statistics
- Bivariate/multivariate analysis: relationships between important variables
- Target/class distribution
- Correlation or association analysis where meaningful
- Relevant trends, patterns, anomalies or clusters
- EDA-based observations that influence preprocessing or modelling decisions

6) Model Development Requirements

- Establish a baseline before presenting the final model.

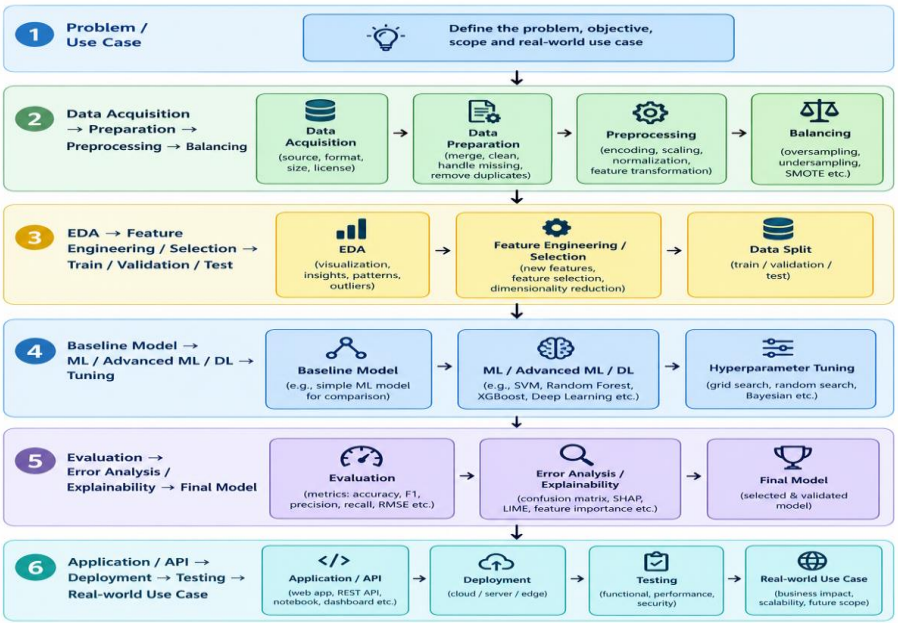
- Use at least two relevant modelling approaches when the problem permits meaningful comparison.
- Explain the assumptions and suitability of each selected method.
- Tune important hyperparameters rather than relying blindly on default settings.
- Use appropriate validation or cross-validation.
- Report training and validation/test performance separately where relevant.
- For Deep Learning, document architecture, activation functions, optimizer, loss function, epochs, batch size and regularization choices.
- Do not claim improvement solely because a model has a higher training score.

7) Evaluation & Validation

Problem Type	Suggested Metrics	Additional Analysis
Classification	Accuracy, Precision, Recall, F1-score	Confusion matrix, ROC-AUC/PR-AUC where appropriate, class-wise results
Imbalanced Classification	F1, Recall, Precision, PR-AUC, balanced accuracy	Minority-class performance and threshold considerations
Regression	MAE, MSE/RMSE, R^2	Residual/error analysis and actual-vs-predicted analysis
Clustering	Silhouette score and other suitable measures	Cluster interpretation and visualization
Computer Vision / DL	Task-appropriate metrics	Class-wise errors, sample predictions and failure cases
NLP	Task-appropriate metrics	Error examples and qualitative analysis

Students must explain why the selected metric(s) are appropriate. Accuracy alone is generally insufficient when class distributions are uneven.

8) Recommended Project Architecture



9) Academic Integrity & Use of Generative AI

Students may use AI-assisted tools for brainstorming, coding support, debugging, documentation or learning where permitted by course policy. However, students remain responsible for understanding, verifying and being able to explain every part of their project. Copied code, fabricated results, fabricated references, manipulated evaluation results or a project that the team cannot explain may be treated as academic misconduct.

- Maintain a record of major datasets, code sources, libraries and external resources used.
- Verify AI-generated code before using it.
- Do not fabricate experimental results or claim deployment that was not actually demonstrated.
- Be prepared for individual technical questions during evaluation.

Key Message: A successful ML project is not merely a model with a high accuracy. It is a justified, validated and reproducible solution to a meaningful problem, developed through a complete data-to-deployment pipeline.